Renewable Generation Forecasting

Case Study

Scene Setting

This case study looks at how Open Climate Fix uses open-source artificial intelligence to improve renewable energy forecasting in the electricity grid. Faced by the growing constraints of intermittent wind and solar, grid operators must balance electricity supply and demand. Through collaboration with Google DeepMind, they aim to create accurate and reliable renewable forecasting that can help grid operators with this complex task, therefore reducing costs and emissions.

Their work is a demonstration of how a shift from solely relying on numerical weather predictions to integrating AI in renewable forecasting can significantly improve the accuracy and speed of renewable energy output predictions - crucial for efficient grid operations. Better renewable forecasts lead to reduced waste generation, lower costs, lower emissions, and increased grid reliability.

Background Context

Open Climate Fix is a non-profit organization focused on using open-source AI and machine learning with the goals of achieving lower electricity costs and reducing carbon emissions. They built AI tools to improve tasks such as solar, wind, and cloud forecasts, which are beneficial to grid operators in their day-to-day operations. They make their work accessible through GitHub and Hugging Face to encourage collaboration from researchers working on solutions for a smart energy grid. Open Climate Fix works with energy providers, grid operators, traders, tech innovators, and researchers. In 2025, Open Climate Fix partnered with Google DeepMind to enable grid operators to use AI weather forecasts. The mission of Open Climate Fix is becoming especially important in the context of today's energy industry, where grid operators are facing growing constraints.

Electricity grids are rapidly adopting renewable generation as they shift towards cleaner energy and work towards reaching emissions goals. However, an increasing reliance on variable wind and solar presents challenges for grid operators, as these energy sources are not always readily available. These constraints make balancing power supply and demand more difficult, therefore requiring reliable short-term renewable forecasts.

Main Issue

As energy grids transition to cleaner energy, more systems are adopting renewable energy; however, these sources are intermittent, meaning they are not always available when we need them. Grid operators are constrained by the intermittency of renewable energy resources. The constraint makes the jobs of the grid operators more difficult because the weather has an increasing impact on available energy. Grid operators must have access to other forms of energy as backups when needed. But how do they know when other, non-renewable energy sources, such as oil and gas, will be needed?

> **"Our electricity grids are struggling to keep up with the shift to renewables, leading to grid instability, excess carbon and higher costs."**
>
> - Open Climate Fix Problem Statement

When predicting future renewable generator output, grid operators rely on weather forecasts. Historically, this was done by Numerical Weather Predictions (NWPs), which is a type of forecasting that uses mathematical models of the atmosphere to predict the weather based on current conditions (Open Climate Fix, 2025). However, NWPs are slow and can be fragile to changing conditions.

There was an opportunity for weather forecasting to be improved, to save time and computing costs. Better weather forecasts from using AI can help grid operators manage the challenge. AI prediction models could generate ensembles more quickly and reduce computing costs as compared to NWPs. Additionally, AI weather prediction models could predict more days forward, meaning that operators could have more time to secure backup electricity supply from non-renewable resources.

The AI Angle

AI Context & Significance

AI was introduced to bridge the gap between NWPs and optimal decision-making. It helped to translate detailed weather forecasts into accurate wind and solar predictions. NWPs produce detailed atmospheric variables such as cloud cover and wind speed, but they do not directly translate those conditions into site-specific estimates of wind and solar generation. AI bridges the gap by learning how those complex weather patterns result in renewable energy output. Rather than replacing numerical weather predictions, Quartz Solar uses NQP outputs as inputs, learning how atmospheric variables translate into site-specific generation patterns. Grid operators use these predictions to make energy decisions such as scheduling power from different sources, curtailing renewable output, and balancing supply and demand.

SOLAR FORECASTING

Traditional weather forecasting models are slow to provide updates, and typically focus on broader weather patterns as opposed to a specific area. On the other hand, AI can be used for solar and cloud forecasting by combining different data, such as satellite imagery, numerical weather predictions, topographic data, and real-time PV readings, to produce accurate forecasts of solar generation. This helps grid operators balance supply and demand more efficiently and reduces reliance on fossil-fuel reserve generation.

WIND FORECASTING

Wind energy is difficult to predict, as it can be produced at any time of the day; there are more factors that can impact wind generation as opposed to solar. Because grid operators need to reduce uncertainty, they need accurate wind forecasts to keep the grid stable. Being able to accurately predict wind generation can result in less curtailment and ultimately lower energy costs for consumers.

Open Climate Fix uses AI for wind forecasting because traditional methods struggle to capture complex patterns in wind generation, which depend on many interacting variables and geographic factors.

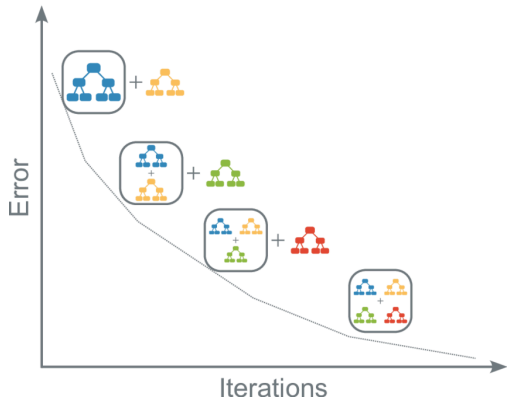
Open Climate Fix Forecasting Initiatives		
Task	**Model**	**Case Study**
Wind Forecasting	Quartz Wind	RUVNL, Rajasthan, India
Solar Forecasting	Quartz Solar	NESO, United Kingdom

AI TechniquesOpen-source Quartz Solar forecasting model uses machine learning models that map weather forecast inputs to solar power output predictions at a single site.

There are model types for forecasting that are used:

1. Gradient Boosting Model
2. XGBoost Model

Gradient Boosting Machine (GBM)



A gradient-boosting model is a machine learning method that predicts by combining the outputs of multiple decision trees. Each new tree focused on correcting the mistakes of the previous ones to reduce errors. Instead of trying to learn everything at once, this model learns step by step, gradually improving its accuracy. The correction process makes GBMs good at modeling complex nonlinear relationships. For example, how certain weather variables affect solar output.

Figure 1. Visualization of GBM (Zuchniak, 2023).

The primary Quartz Solar model is a GBM. The model is trained on 25,000 solar sites using 5 years of data. The training is for the model to learn how the weather influences the solar output. Gradient boosting models are

particularly effective for structured tabular data such as weather variables, where nonlinear interactions between variables significantly influence output generation.

It uses 9 weather prediction variables:

1. Visibility (km)
2. Wind Speed at 10 meters (km/h)
3. Temperature at 2 meters (°C)
4. Precipitation (mm)
5. Shortwave Radiation (W/m^2)
6. Directional radiation (W/m^2)
7. Cloud Cover low (%)
8. Cloud Cover mid (%)
9. Cloud Cover high (%)

Additionally, it uses some engineered variables.

eXtreme Gradient Boosting (XGBoost)

XGBoost is a more advanced and optimized version of gradient boosting. Like standard gradient boosting, it builds many small decision trees, with each new tree correcting the errors of the previous.

However, it is faster as it uses parallel processing for speed, built-in L1 and L2 regularization to reduce overfitting, and handles missing values. It is a more complex model than GBM; there is a tradeoff between accuracy and transparency. The alternative model available is an XGBoost model using slightly different weather features from open-meteo data.

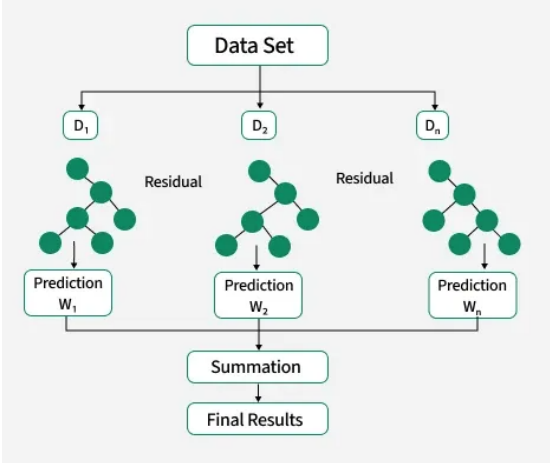


Figure 2. Visualization of XGBoost (GeeksforGeeks, 2025).

Model Comparison

A comparison of the models is displayed in the figure below. The differences are due to the selected models and input data, showing that predictions can vary slightly across locations but still share a similar overall shape.

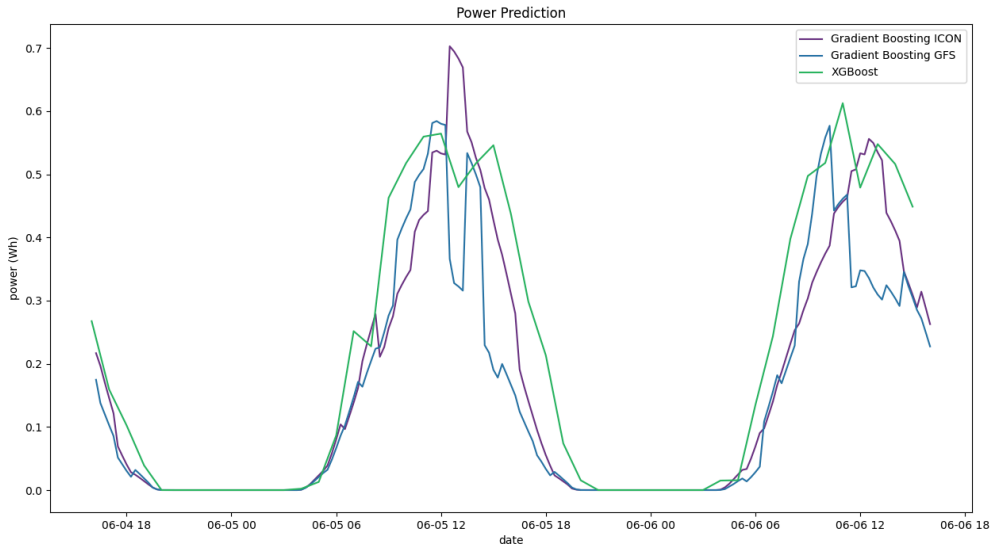


Figure 3. Power prediction comparison of different models (Open Climate Fix, 2025).

Plot Twist

Solar UK Case

The UK's National Energy System Operator (NESO) had seen that solar generation had been growing rapidly, and they identified it as one of their key sources of uncertainty. So, Open Climate Fix worked with NESO to integrate their Quartz Solar service. This model is used by control room engineers for real-time balancing, dynamic reserve setting, and NESO total system demand forecasts.

The results:

- 300, 000 tonnes reduction in CO2 emissions
- £30 million annual cost savings

These savings were attributed to improved reserve scheduling and reduced fossil-fuel balancing activation due to more accurate short-term solar forecasts.

India Wind Case

The state of Rajasthan has the highest renewable capacity in all of India, including many GW of wind generation. Open Climate Fix is forecasting the wind generation both day ahead and intra-day.

While this is a newer case study, Open Climate Fix states that the impact for operators in India is:

- 30% increase in forecasting accuracy
- 100% reduction in forecasting penalties

Open Climate Fix reports a 30% improvement in forecasting accuracy compared to previous baseline models used by operators in Rajasthan.

Data and Deployment Limitation

1. Different data sources could lead to inconsistencies

Restriction: Different NWP data sources are used depending on whether the predictions are within the last 90 days or further in the past.

What was noticed: The same model behaves differently depending on when a forecast is made. This is due to underlying weather inputs (different formats and availability).

Result: The forecast quality also depends on the data availability constraints, making it harder to test historically as opposed to live forecasts.

2. Training and inference data mismatch can lead to odd behaviour

Restriction: The model was trained on UK Met Office NWP data, but when inference is run on GFS data from open-meteo.

What was noticed: Although the weather variables are almost the same, differences in how force-acting agencies model the atmosphere can affect the relationships learnt by the model during training.

Result: Using a different weather model at inference can lead to odd behaviours in predictions. This shows that AI performance depends on alignment between the training and deployment data sources.

So, What?

- AI renewable forecasting improves the accuracy and speed of renewable energy generation predictions.

- More reliable short-term forecasts support grid operators in their daily decision-making by helping them balance electricity supply and demand with variable renewable generation.

Open Climate Fix's models improve predictions of solar and wind generation. This forecasting task helps grid operators in their day-to-day work by providing accurate renewable generation forecasts more quickly. More accurate and timely forecast predictions support operators in making better decisions while balancing electricity supply and demand. The value of tools, such as AI for renewable resource forecasting, is not fully realized without grid operators and human judgment. More broadly, this approach shows that as electricity grids incorporate higher levels of renewable energy, AI will play an important role in assisting grid operators, reducing costs, and lowering emissions; not by replacing human operators, but by helping to shape how decisions are made in increasingly complex energy systems.

Implications & Considerations

The use of AI in renewable forecasting helps shift grid operations from being reactive to proactive with variable renewable generation. Accurate predictions can help operators plan ahead and be prepared to have non-renewable backup generation when needed.

Operational Considerations

- **Human judgment remains important** in operational use. AI renewable forecasts can improve accuracy, but they do not fully replace the human decision-making done by grid operators.

- **Data matters as much as the model.** The effectiveness of AI forecasting tools depends heavily on the input data. Differences in training and deployment data can affect model behaviour.

This highlights the importance of data governance and model monitoring when deploying AI forecasting tools in operational environments.

Let's Talk

1. AI forecasting can sway grid operators to make decisions in a certain way. How is responsibility shared when decisions are made on inaccurate forecast predictions?

References

Konrad Zuchniak (2023, February). Illustration of a gradient boosting machine. *https://www.researchgate.net/figure/Illustration-of-a-Gradient-Boosting-Machine_fig3_368507910*

GeeksforGeeks. (2025, September 5). *Implementation of XGBoost (Extreme Gradient Boosting).* *https://www.geeksforgeeks.org/machine-learning/implementation-of-xgboost-extreme-gradient-boosting/*

Open Climate Fix (2025, September 12). README-open-source-quartz-solar-forecast. *https://github.com/openclimatefix/open-source-quartz-solar-forecast/blob/main/README.md*

Open Climate Fix (n.d.). https://www.openclimatefix.org/

amii.ca | hello@amii.ca

Alberta Machine Intelligence Institute
1101, 10065 Jasper Ave NW
Edmonton, AB T5J 3B1 Canada